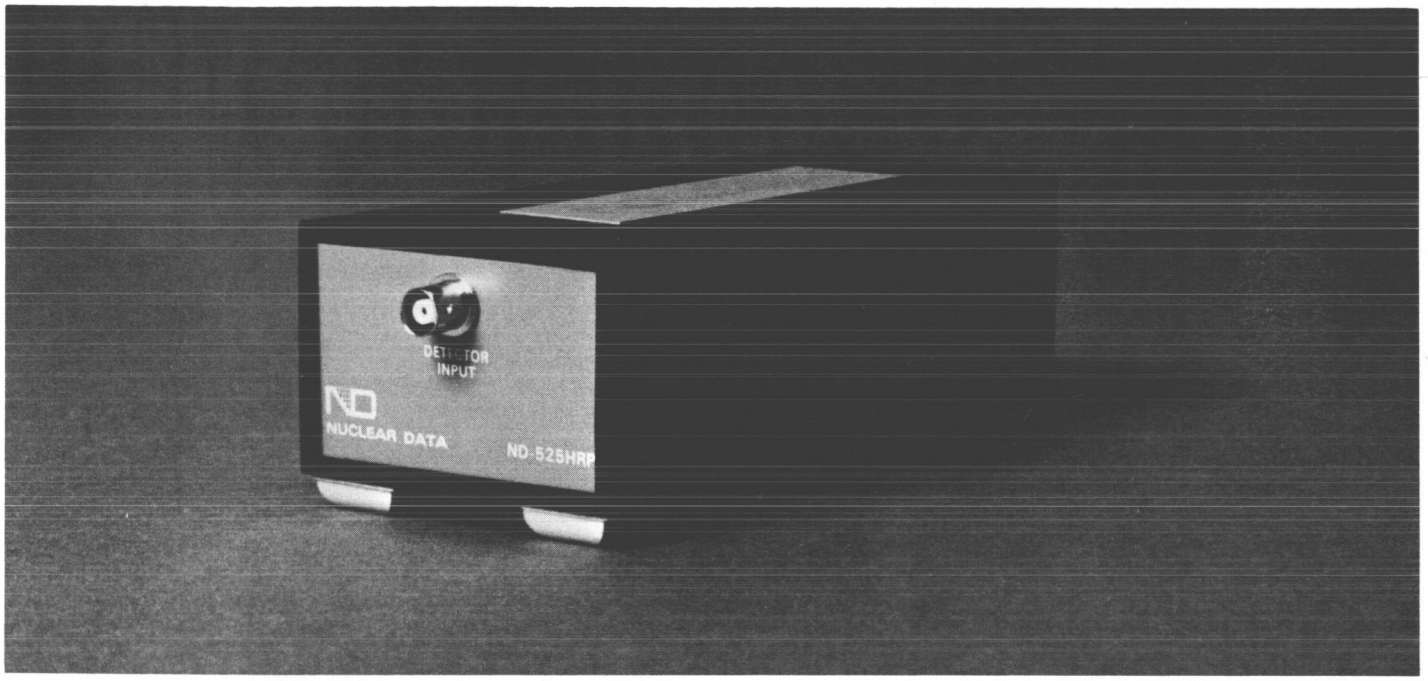
TECHNICAL SPECIFICATIONS

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ND-525HRP HIGH RESOLUTION PREAMPLIFIER

Extremely high resolution capabilities inherent in cooled Ge(Li) detectors are realized only to the extent of limits imposed by associated electronics, especially the necessary charge-sensitive preamplifier. The ND-525HRP is a high resolution charge sensitive preamplifier which provides the high degree of performance consistent with state-of-the-art technique.

Two selected field effect transistors (FET's), connected in parallel, are used in the input circuitry to provide excellent low noise performance. The ND-525 can also be supplied, at additional cost, with four parallel-connected FET's to provide greater low noise performance with detector capacitance exceeding 250 picofarads. A decay time constant of 55 microseconds, which is pole-zero compensated to minimize pulse undershoot effects, is provided by the charge-sensitive circuit group. The voltage output from the charge-sensitive group is amplified by a factor of four voltage sensitive group. A single component in this group can be changed to yield any amplification factor from 2 to 16, if desired.

Operating voltages can be obtained from the companion ND-524LARC linear pulse amplifier preamplifier power output, or from a standard AEC bin power supply.

CHARGE SENSITIVE GROUP

Conversion Gain:

54 mv/Mev (Ge).

Input Polarity:

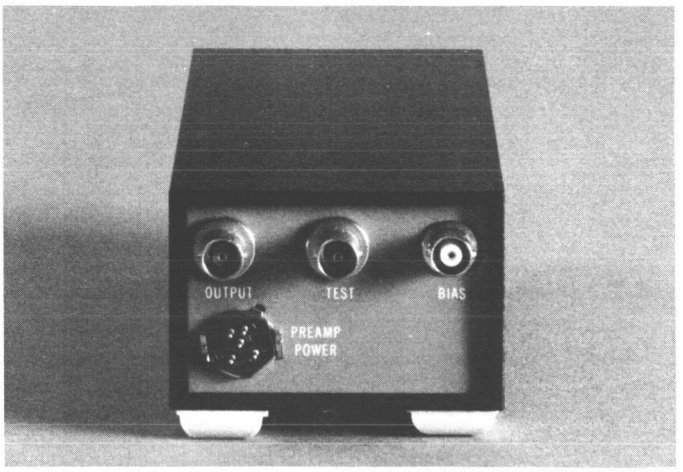
Either.

Noise (Ge):

Less than 2 Kev at 10 pf, external capacitance.

Noise Slope:

Less than .04 Kev/pf, external capacitance.



ND-525HRP High Resolution Preamplifier rear panel showing Output, Test, Bias, and Preamp Power connectors.

Input Capacitance:
Hot — 20,000 pf (minimum).
Cold — 20 pf.

Rise Time:
50×10^{-9} sec at 0 pf, external capacitance.
80×10^{-9} sec at 100 pf, external capacitance.
150×10^{-9} sec at 1000 pf, external capacitance.

Sensitivity Loss as a Function of Detector Capacitance:
2% at 500 pf.
5% at 1000 pf.

Decay Time Constant:
55 microseconds ±2%.

VOLTAGE SENSITIVE GROUP

Voltage Gain:
x4; standard (single) component change gives any value from 2 to 16.

Output Impedance:
100 ohms ±1% D.C. coupled.

Output Voltage:
±10 volts linear; ±12 volts saturated into open circuit.
±5 volts linear; ±6 volts saturated into 100 ohms.

GENERAL

Output Polarity:
Inversion of input.

Integral Linearity:
Less than 0.1% for 0-10 volt output (unterminated).

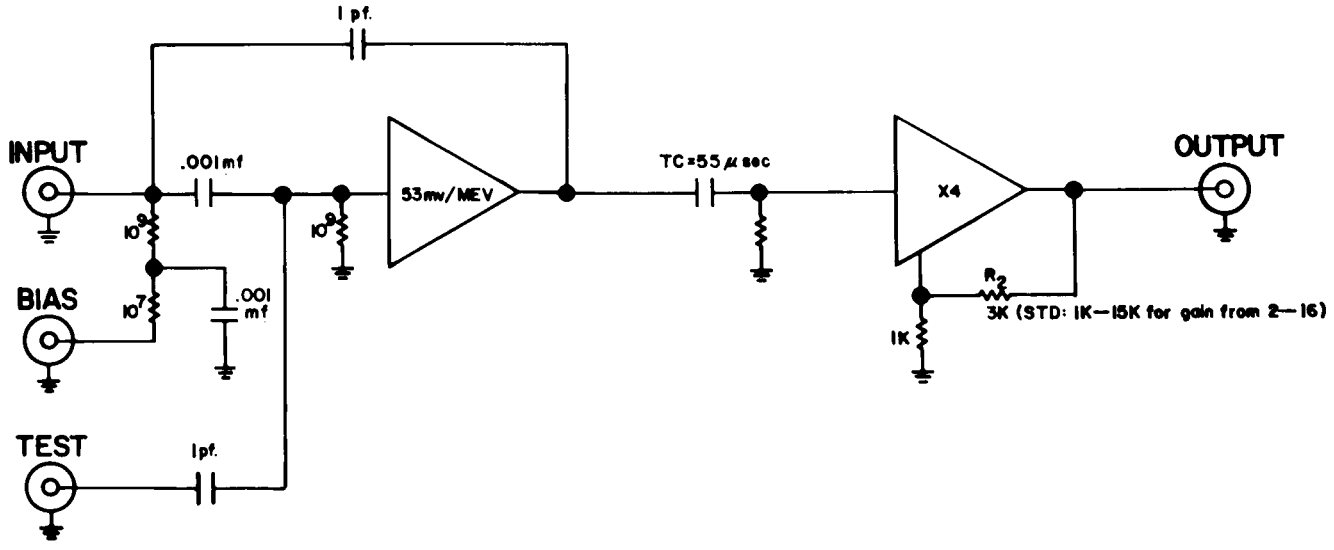
Temperature Stability:
0.005%/°C.

Connectors:
Input - UG 931/U (MHV).
Detector Bias - UG 931/U (MHV); Up to 2000 volts bias may be applied.
Test - UG 290/U (BNC); An internal 1.0 pf ±.25 pf test capacitor is provided.
Output - UG 290/U (BNC).
Power - Amphenol 126-218.

Power Requirements:
+24 volts at 20 ma.
−24 volts at 20 ma.
+12 volts at 30 ma.

Size:
7-1/8" long x 3" wide x 2" high, exclusive of connectors and mounting feet.

Weight:
1.25 pounds.



BLOCK DIAGRAM OF ND—525 HRP

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